CMPG 311

GROUP PROJECT DATABASE INITIAL STUDY (PHASE 1)

GROUP 14

PROJECT 16: VEHICLE RENTAL SYSTEM (DriveFlow Rentals)

25 March 2026

Table of Contents

Table of Contents 2
1. Group Information 3
2. Company Situation Analysis 5
2.1 Mission and Objectives 5
2.2 Operational Description 5
2.3 Organisational Structure 5
3. Problems and Constraints 6
3.1 Identified Problems 6
3.2 System Constraints 6
4. Database System Specifications 7
4.1 Objectives 7
4.2 Proposed Database Schema 7
4.3 Information Requirements 7
4.4 Scope Definition 8
4.5 Boundary Definition (Out of Scope for Phase 1) 8
5. References 9

Contribution Table

Name and surname	Contribution Status
1. Kudzai Mudzingwa	Contributed
2. Xoliswa Sentse	Contributed
3. Trevour Sithole	Contributed
4. Edition Baloyi	Contributed
5. Shekinah Khoza	Contributed
6. Thato Mandelisa	Contributed
7. Rifuwo Mbombi	Contributed

1. Group Information

1.

Project Manager & Lead Architect
Kudzai Mudzingwa
44177348@mynwu.ac.za
+27 76 319 0045

2.

Database Administrator
Xoliswa Sentse
37940066@mynwu.ac.za
+27 81 459 5244

3.

Backend Developer (Core Logic)
Trevour Sithole
45485097@mynwu.ac.za
+27 69 161 6943

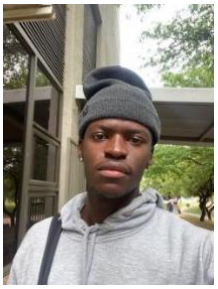
4.

Backend Developer (Integrations)
Edition Baloyi
43796540@mynwu.ac.za
+27 79 300 2793

5.

Frontend Developer (User Interface)
Shekinah Khoza
44833563@mynwu.ac.za
+27 68 252 3359

6.



Frontend Developer (Admin Dashboard)
Thato Mandelisa
34286799@mynwu.ac.za
+27 69 110 2561

7.

Quality Assurance & Documentation
Rifuwo Mbombi
40676099@mynwu.ac.za
+27 66 539 1680

2. Company Situation Analysis

2.1 Mission and Objectives

DriveFlow Rentals' mission is to provide customers with a highly-efficient and digital vehicle rental and procurement solution. DriveFlow Rentals aims to replace the use of paper-only tracking systems with a centralised database that ensures the 100% integrity of the company's data. Through its platform, DriveFlow Rentals aims to provide customers with an easy to use, reliable and fully digital rental experience for their vehicles.

2.2 Operational Description

DriveFlow Rentals has a fleet of cars that it owns and maintains. Cars are procured, registered, insured, and maintained to ensure that they are always in great running condition. The company has an online and physical presence where customers can view available cars and make bookings for them. The company manages car bookings to ensure that there is no conflict in the booking of cars by customers and to maximize the utilization of the company's fleet.

Upon pickup, customers are required to present themselves and produce identification to verify their identity, license, and payment details, after which they are issued a rental agreement. During the rental period, the car rental company may monitor the customer's cars or provide support to the customer as necessary. Upon return to the company, the cars are inspected for damage, mileage, and fuel levels, are cleaned and maintained, and then returned to the available fleet for rental while processing the customer's rental billing statement.

2.3 Organisational Structure

DriveFlow Rentals operates with a structured hierarchy divided into three functional levels, each mapped to specific system access roles:

Level	Roles	System Access
Administrative	Senior Management	Full system access: pricing, fleet acquisition, financial planning, compliance, system audits
Operational	Fleet Manager, Coordinators, Technical Team	Fleet management, vehicle procurement, inspection scheduling, database management, reporting
Customer Service	Customer Representatives	Reservations, booking modifications, dispute resolution, manual overrides

3. Problems and Constraints

3.1 Identified Problems

The following distinct operational problems have been identified within the current manual system, listed in order of criticality:

Priority	Problem	Description
Critical	Double Booking / Concurrency Issues	The manual system lacks real-time updates and concurrency control, allowing two customers to book the same vehicle simultaneously.
Critical	Data Fragmentation	Customer history is stored across separate, unlinked files, making it impossible to identify high-value repeat customers or build loyalty programs.
High	Reporting Latency & Errors	Monthly revenue requires manual tallying of paper receipts, which is both time-consuming and prone to human error.

3.2 System Constraints

Category	Constraint
Technological	The solution must be developed using the Flask (Python) framework with an HTML/CSS frontend and a relational database backend (PostgreSQL/MySQL).
Security	The system must implement password hashing (Bcrypt/Werkzeug) to protect user credentials.
Legal	Sensitive personal data (e.g., driver's licence numbers) must be stored securely in compliance with the Protection of Personal Information Act (POPIA), Act 4 of 2013.
Scope	The system must support up to 100 concurrent users and manage a fleet of up to 500 vehicles.
Time	The initial database implementation and study must be submitted by the specified module deadline within the current semester.

4. Database System Specifications

4.1 Objectives

The following database objectives will help to address the problems described in Section 3:

- Ensure that double-bookings of vehicles are eliminated by implementing UNIQUE constraints within the vehicle schedules table.
- Ensure that the database is fully normalised to enable the tracking of customer rental histories.
- Utilise SQL aggregate functions to automatically create rental reports for the business, eliminating the necessity of manually calculating the rental revenues of the vehicle hire business.

.

4.2 Proposed Database Schema

The database will consist of the following core entities and their attributes:

Table	Key Attributes	Notes
Users	user_id (PK), full_name, email (UNIQUE), licence_number (encrypted), password_hash, created_at	Stores customer and admin accounts. Licence numbers encrypted at rest per POPIA.
Vehicles	vehicle_id (PK), vin (UNIQUE), make, model, year, daily_rate, status, branch_id (FK)	Status values: Available, Rented, Under Maintenance.
Bookings	booking_id (PK), user_id (FK), vehicle_id (FK), pickup_date, return_date, total_cost, payment_status	UNIQUE constraint on (vehicle_id, pickup_date, return_date) prevents double-booking.
Branches	branch_id (PK), branch_name, address, city	Supports multi-location fleet management.
Maintenance	maintenance_id (PK), vehicle_id (FK), service_date, description, next_due_date	Tracks vehicle service history.

4.3 Information Requirements

The database will need to capture and relate the following data points:

- Users will include fields for their full name, email address, password, and licence number fields that are encrypted and stored in the database.

- Vehicles will include fields for the vehicle's VIN, make, model, year, daily rental rate, availability status fields, and the branch where the vehicle is assigned.
- Bookings will include fields for the rental's pickup and return date, total cost for the rental, payment status fields, and fields to associate with both the user and vehicle in the database.
- Branches will include fields for the branch's name, address, and the city in which the branch is located.
- Maintenance Records will include fields for the service date for the vehicle, a description of the maintenance performed, and when the next service is due for the vehicle.

4.4 Scope Definition

The features of the system will include:

- The number of users that will be supported concurrently will be 100
- The number of vehicles that will be supported within the system will be 500
- These limitations are based upon the initial deployment of the system to a limited number of branches

4.5 Boundary Definition (Out of Scope for Phase 1)

The following features will NOT be included in Phase 1, but which could be incorporated into future phases of the project:

- GPS data processing and live tracking of vehicles
- Insurance claims integration
- Integration with real-world bank APIs (simulated during Phase 1)
- Implementation of an email or SMS notification system
- Mobile application development

5. References

Connolly, T.M. and Begg, C.E., 2015. Database Systems: A Practical Approach to Design, Implementation, and Management. 6th ed. Harlow: Pearson.

Pallets Projects, 2024. Flask Documentation (v3.x). [online] Available at: https://flask.palletsprojects.com/ [Accessed 25 March 2026].

PostgreSQL Global Development Group, 2024. PostgreSQL 16 Documentation. [online] Available at: https://www.postgresql.org/docs/ [Accessed 25 March 2026].

SQLAlchemy, 2024. SQLAlchemy Documentation. [online] Available at: https://www.sqlalchemy.org/ [Accessed 25 March 2026].

Republic of South Africa, 2013. Protection of Personal Information Act, No. 4 of 2013. Pretoria: Government Gazette.